

		Parul University Faculty of Engineering and Technology Parul Institute of Engineering and Technology Robotics & Artificial Intelligence Department	
Subject Name	Computer Organization and Microprocessor (COMA)	A.Y	2025-26
Subject Code	303105210	Semester	4th

Chapter-1			
Sr No	Question	COs	B.T
1	Draw and explain the programming model of 8085.	CO1	BT-2
2	Define the following (1) Machine Language (2) Assembly Language (3) Assembler (4) subroutines	CO1	BT-1
3	Draw and explain Architecture of 8085 Microprocessor.	CO1	BT-2
4	Give difference between microprocessor and microcontroller.	CO1	BT-2
5	Explain direct addressing mode and Indirect addressing mode of 8085 with examples.	CO1	BT-2
6	Draw pin diagram of 8085 microprocessor and explain READY and RESET IN pin.	CO1	BT-2
7	Draw pin diagram of 8085 and explain any 5 pins in detail	CO1	BT-2
8	Explain flag register in detail.	CO1	BT-2
9	What is the use of register in 8085? How register is different from memory?	CO1	BT-2
10	Draw and explain de-multiplexed address data bus.	CO1	BT-2

11	Draw and explain any three blocks of architecture of 8085.	CO1	BT-2
12	Classification of Microprocessor based on size, application and speed.	CO1	BT-4
13	Explain languages for communication between microprocessor to other peripheral and vice-versa.	CO1	BT-2
14	Explain the need to De-multiplex the bus AD7-AD0.	CO1	BT-2
15	Define: bit, nibble, byte, word and instruction	CO1	BT-1
16	What is meaning by 8-bit processor?	CO1	BT-2
17	What is Mnemonics?	CO1	BT-1
18	Define: Assembly language, machine language, high level language	CO1	BT-1
19	What is a Compiler?	CO1	BT-1
20	Which information given by address line?	CO1	BT-1
21	Data bus is always bidirectional True/False	CO1	BT-1
22	Microcontroller is faster than Microprocessor justify the statement.	CO1	BT-2
23	What are the advantages of an assembly language in comparison with high level language?	CO1	BT-2
24	Write full name of: RAM, ROM, ALU, SSI, MSI, VLSI, and ASCII.	CO1	BT-1
25	What is bus?	CO1	BT-1
26	How much memory access by 8085 microprocessor.	CO1	BT-1

27	ALU stand for.	CO1	BT-1
28	What is purpose of Control Unit of Microprocessor?	CO1	BT-2
29	Justify “ Microprocessor is a programmable device”	CO1	BT-2
30	Justify “The direction of address bus is unidirectional”	CO1	BT-2
31	What is difference between 4 bit microprocessor and 8 bit microprocessor?	CO1	BT-2
32	What is difference between assembly language and machine language?	CO1	BT-2
33	What is use of Registers in microprocessor?	CO1	BT-2
34	Define: Opcode and Operand.	CO1	BT-1
35	What is difference between general purpose microprocessor and special purpose microprocessor?	CO1	BT-2
36	Write any one three byte instruction of 8085	CO2	BT-1
37	_____ pin is indicates that another device is requesting the use of the address and data bus of 8085	CO1	BT-1
38	ASCII code stands for_____.	CO1	BT-1
39	How many transistors are used in 8085 microprocessor?	CO1	BT-1
40	Flag register is of _____ bit.	CO1	BT-1

Chapter-2

Sr No	Question	COs	B.T
1	Draw and explain the Interfacing of 4KB EPROM to 8085 with starting address A000H.write the starting and ending location of 8085	CO3	BT-6
2	Draw memory interfacing of EPROM of 2Kbyte and RAM of 4Kbyte with 8085 using the decoder.	CO3	BT-6
3	Interface two memories RAM and EPROM having size of 8kbyte each. Find its starting address and Ending address.	CO3	BT-6
4	What do mean by peripherals?	CO3	BT-1
5	Can we interface 64 MB memory with 8085?(Y/N)	CO3	BT-4

Chapter-3

Sr No	Question	COs	B.T
1	Explain CMP M , RAL and CMA instruction of 8085	CO2	BT-2
2	Explain given Instructions: 1) MVI 2) CMP 3) XRA 4) SUB 5) STA	CO2	BT-2
3	Describe following instruction in detail with example: I. LXI instruction II. LHLD III. STAX IV. CMA V. RRC	CO2	BT-2
4	Write a program to divide two 8-bit data given as per memory location, Input: 7015H – 50H 7016H – 05H Store output at 5016H memory location.	CO2	BT-3
5	For 8085 microprocessor, if register B has 65 H and accumulator has 97 H, What will be the content of accumulator after executing instruction SUB B?	CO4	BT-4
6	Explain usage of STAX instruction.	CO2	BT-2

Chapter-4			
Sr No	Question	COs	B.T
1	Generate 0.5second delay using 16-bit register pair as counter method. Consider 6MHz clock frequency and the internal clock frequency is 3MHz.	CO2	BT-3
2	Write 8085 Assembly language program to convert binary to ASCII values.	CO2	BT-3
3	Explain the stack and subroutine with role of PUSH and POP instructions.	CO2	BT-2

Chapter-5			
Sr No	Question	COs	B.T
1	Draw and explain interrupt structure of 8085	CO3	BT-2
2	Explain Vectored, Non vectored, Maskable and Non Maskable interrupts. Give examples also.	CO3	BT-2
3	Classify Interrupts and explain it with example.	CO3	BT-4
4	Explain RIM and SIM instruction in detail	CO3	BT-2
5	Which instruction is used to control interrupt. Give its types and explain any one with diagram.	CO3	BT-2
6	Describe what is an Interrupt? Explain each priority wise Interrupt types. Also describe EI and DI instruction?	CO3	BT-2
7	In 8085, Which of the following instruction use to check interrupts are pending or not.	CO3	BT-1

Chapter-6

Sr No	Question	COs	B.T
1	Explain logical and circular shift microoperation	CO1	BT-2
2	Explain the fetch stage; decode stage and execution stage of instruction cycle with example of any one instruction.	CO1	BT-2
3	Explain circular shift and arithmetic shift micro operations with examples.	CO1	BT-2
4	Draw timing diagram of instruction MOV B, C.	CO1	BT-4
5	Draw a common bus system using multiplexers.	CO1	BT-6
6	Explain MAR and MAD with memory diagram.	CO1	BT-2
7	Draw and explain timing diagram for MVI B, 05H.	CO1	BT-4
8	What is the instruction and how it will be fetch and executed?	CO1	BT-2
9	Define Machine cycle. List its types and draw Timing diagram for IO read Instruction.	CO1	BT-4
10	What is an Assembler?	CO1	BT-1
11	Why do we need clock in microprocessor?	CO1	BT-2
12	What stand for CISC and RISC?	CO1	BT-1
13	What is MIMD and SIMD?	CO1	BT-1

Chapter-7			
Sr No	Question	COs	B.T
1	Give classification of memory in 8085.	CO1	BT-4
2	Elaborate cache memory.	CO1	BT-2
3	What is the high-speed memory between the main memory and the CPU called?	CO1	BT-1
4	Flash Memory is non-volatile memory? (Y/N) Justify your answer.	CO1	BT-2